[INCH-POUND]
OO-L-2880
August 17, 1997
(SUPERSEDING)
MIL-L-9223A(YD)
31 January 1989

FEDERAL SPECIFICATION

LAUNDRY UNIT, SKID-MOUNTED, 100 POUNDS (DRY WEIGHT) PER HOUR

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type and size field laundry unit assembled on a 7 foot (2.13 meters (m)) by 9.5 foot (2.89 m) skid, with a minimum capacity of not less than 100 pounds (lb) (45.34 kilograms (kg)) (dry weight) of laundry per hour.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification

OO-D-1297 - Drying and Conditioning Tumblers, Laundry, Commercial: General Specification For

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be addressed to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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Federal Standards

FED-STD-595	-	Colors Used In Government Procurement
FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-H28/7	-	Screw-Thread Standards For Federal Services Section 7 Pipe Threads - General Purpose
FED-STD-H28/10	-	Screw-Thread Standards For Federal Services Section 10 Hose Coupling And Fire Hose Coupling Screw Threads

Commercial Item Description

A-A-1106	-	Extinguisher, Fire, Carbon Dioxide

Military Specifications

MIL-B-3180	-	Boilers and Related Equipment, Packaging Of
MIL-H-16451	-	Heaters, Water, Liquid Fuel, (Oil-Fired, Storage Type: Not Exceeding 280,000 Btu/hr Input Capacity)

Military Standards

MIL-STD-461	-	Control Of Electromagnetic Interference Emissions And Susceptibility, Requirements For The
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of

(Activities outside the Federal Government may obtain copies of federal specifications and standards as specified in the General Information section of the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index is for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402)

(Single copies of the specification, and other federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.)

(Federal Government activities may obtain copies of federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

(Copies of federal and military specifications and standards required by contractors in connection with specific procurement functions may be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue in effect on the date of invitation for bids or request for proposal shall apply, unless a specific issue is identified.

DEPARTMENT OF LABOR (DOL)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29 - Environment
Part 1910 - Occupational Safety and Health Standards

(The Occupational Safety and Health Standards are for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for its issuance.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z8.1 - Commercial Laundry and Dry-Cleaning Operations

(Private sector and civil agencies may purchase copies of this voluntary standard from the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.)

ASTM

ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip
ASTM A 494 - Castings, Nickel and Nickel Alloy
ASTM B 62 - Composition Bronze or Ounce Metal Castings
ASTM B 127 - Nickel-Copper Alloy (UNS N04400) Plate, Sheet and Strip
ASTM B 164 - Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B 564 - Nickel Alloy Forgings
ASTM B 584 - Copper Alloy Sand Castings for General Applications
ASTM D 3951 - Commercial Packaging

(Private sector and civil agencies may purchase copies of these voluntary standards from the ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 2 - Industrial Control and Systems Controllers, Contractors, and Overload Relays Rated Not More Than 2000 Volts AC or 750 Volts DC
NEMA ICS 6 - Industrial Control and Systems Enclosures
NEMA MG 1 - Motors and Generators
NEMA WC 5 -Thermoplastic-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Private sector and civil agencies may purchase copies of this voluntary standard from the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J534 - Lubrication Fittings

(Private sector and civil agencies may purchase copies of this voluntary standard from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS LABORATORIES (UL)

UL 67 - Panelboards
UL 489 - Molded-Case Circuit Breakers and Circuit-Breaker Enclosures
UL 1206 - Electric Commercial Clothes-Washing Equipment

(Private sector and civil agencies may purchase copies of these voluntary standards from the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The laundry unit shall be provided with all commercial type equipment, consisting of an open end washer, an extractor, an oil fired dryer, an oil fired hot water heater, a pressurized water system, an electrical panel and other components and accessories specified herein to form an integral unit. The equipment, mounted on a 7.0 foot (2.13 m) by 9.5 foot (2.89 m) skid, shall be arranged for the convenience of the operator. The laundry unit shall be furnished complete, and shall be ready for operation when connected to an outside water, gravity feed fuel oil line, power source, and sewage connection.

3.2 Standard commercial product. The laundry unit shall, as a minimum, be in accordance with the requirements of this specification, and shall be the manufacturer's standard commercial product. Additional or better features that are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the laundry unit being furnished. A standard commercial product is a product that has been sold, or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components, or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated; using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials," means materials that have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.4.1 Corrosion-resistant steel. Corrosion-resistant steel shall conform to any of the 300- or 400-series of ASTM A 167 or A 176, as applicable.

3.4.2 Corrosion-resistant metals. Corrosion-resistant metals used shall conform to the requirements of ASTM A 494 for Nickel and Nickel Alloy Castings; ASTMs B 127, B 164, and B 564 for Nickel-Copper-Alloy; and ASTMs B 584 and B 62 for Copper-Alloy Castings.

3.4.3 Materials for fasteners. Threaded fasteners and rivets shall be of carbon or alloy steel, except where brass or corrosion-resistant metal is to be fastened, in which case they shall be of brass or corrosion-resistant metal respectively (see 3.4.1 and 3.4.2). Where dissimilar metals are fastened, bolts, rivets, screws, and nuts shall be of corrosion-resistant metal (see 3.4.1 and 3.4.2).

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3.4.4 Pipe, tube, fittings, and valves. Pipe, tube, fittings, and valves used for water and fuel oil lines shall be standard commercial items designed for the intended use.

3.4.5 Conduit, fittings, and electrical wire. Electrical conduit shall be rigid, electrical metallic tube or liquid-tight flexible conduit. All electrical wire shall be stranded copper, heat-resistant grade, and thermoplastic insulated conforming to NEMA WC 5.

3.4.6 Bracket, fire extinguisher. A quick release type bracket of the vehicle-type, having a base for floor mounting the extinguisher in an upright position, shall be provided. The bracket shall be sized to accommodate a carbon dioxide hand fire extinguisher conforming to size 10 of Commercial Item Description A-A-1106.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract, shall be identical to the extent necessary to assure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. The laundry unit shall be designed for continuous balance operation as a complete field laundry unit, and for independent operation of each of the components. The laundry unit shall conform to the applicable safety requirements of ANSI Z8.1, Title 29 CFR Part 1910, and as specified herein. Individual components shall be compactly assembled and securely mounted on the skid base specified in 3.7.9. The unit shall be designed for operation on any firm, level, flat surface. The design shall be such as to prevent conditions hazardous to personnel or deleterious to the equipment. Exposed moving or high temperature parts shall be guarded or insulated. Interior surfaces of components coming in contact with the item being laundered shall be free from burrs, sharp edges, snags, or rough areas. Protection to personnel shall be provided in accordance with NFPA 70. All screw threads shall conform to FED-STD-H28.

3.6.1 Dimensions. The laundry unit dimensions shall not be greater than 84 inches in height, 84 inches in width, or 114 inches in length.

3.6.2 Electrical requirements. The laundry units shall be designed for operation on a 208 volt, 3-phase, 60-cycle power supply. Electrical components, including switches and controls, shall be in accordance with NEMA ICS 2.

3.6.2.1 Wiring. Wiring shall conform to the applicable wiring practices, election, and design requirements of NFPA 70 for the operating conditions and load replacements.

3.6.2.2 Power distribution panel. The contractor shall furnish a power distribution panel conforming to UL-67 to provide electrical power to the unit, from the external power source. The power distribution panel shall be a 3 phase, circuit breaker panel rated for not less than 30 amperes at 120 volts or 208 volts. The power distribution panel shall have a main circuit breaker, and a branch circuit breaker for each branch circuit. The power distribution panel shall have a NEMA ICS 6 type 4 enclosure. The power distribution panel shall have a power cable not less than 50 feet in length, to enable connection to an external power source. The power cable shall be a No. 8 American Wire Gage (AWG) portable power cable with grounding conductor, rated

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for 600 volts, and conforming to NEMA WC 5. One end of the cable shall be connected to the incoming (line) terminals of the power distribution panel; the other end shall have 4-inch pigtails for connection to the power source.

3.6.2.3 Motors. Motors protected by unit housings shall be furnished with drip-proof enclosures. Motors not protected by unit housings shall be totally enclosed. Motors shall be constructed, classified and rated in accordance with NEMA MG 1.

3.6.2.4 Circuit breakers and switches. Circuit breakers and switches shall conform to the design and performance requirements of UL-489, and protective requirements of NFPA 70.

3.6.2.5 Motor starters. Each motor over one horsepower shall be furnished with a suitably rated, across-the-line magnetic starter with push-button, on-off station. Magnetic starters shall be constructed, classified, and rated in accordance with NEMA ICS 2.

3.6.2.6 Ground-continuity. Metallic components shall be assembled to ensure ground circuit continuity throughout the complete unit. Suitable copper jumpers shall be used when discontinuities may exist. Electrical grounding of components shall meet the requirements of NFPA 70.

3.7 Construction. The laundry unit shall be complete so that it can be used for the operations specified herein. The laundry unit shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and operation. The water heater and dryer shall be provided with external supports to prevent movement or damage during transportation. In addition, other components and assemblies shall be provided with additional support, if their individual mounting systems are not sufficient to prevent excessive movement or damage during transportation. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted. Warnings shall be mounted on or near components containing hidden hazards. Location of the washing machine, storage rack(s), moisture extractor and dryer on the skid base, shall be such as to permit loading and operation from only two adjacent sides of the skid base.

3.7.1 Washing machine. The washing machine shall be furnished with automatic formula control unit and supply dispenser (see 3.7.1.1), complete with all motor(s) driving mechanisms, starter, controllers, a master switch, timer(s), and other equipment and accessories normally provided with the standard commercial product, for the automatic operation of the washing machine. The washing machine shall conform to the requirements of UL-1206. The washing machine shall be of the end or side loading type, having a laundry washing capacity of not less than 100 lb (45.34 kg) (dry weight), and shall be compatible with the overall laundry unit performance as specified in 3.8. The washing machine shall also be equipped with an automatic thermal cool-down feature for processing permanent press materials. The cylinder and tub of the washing machine shall be fabricated of corrosion-resistant steel.

3.7.1.1 Automatic formula control unit and supply dispenser. The washing machine shall be provided with an automatic formula control unit for directing the machine through a complete

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laundry washing cycle (washing, rinsing) as programmed, and with an automatic supply dispenser for adding supplies as directed by the formula control unit.

3.7.1.1.1 Automatic formula control unit. The unit shall be of the replaceable, plastic, chart-type to automatically control fills, dumps, rinses, and supply injections. The unit shall signal audibly at the end of the cycle, or at other required times. Two charts shall be provided; one pre-cut for a permanent press wash cycle and one uncut spare. The following performance requirements shall apply during laundry operations:

a. The control unit shall automatically control all phases of clutching and braking, and all motor(s).
b. The control unit shall provide automatic control of the drain valve, and the hot and cold fresh water inlet valves, in accordance with programmed formula.
c The control shall automatically control the water level to not less than two predetermined levels which shall be readily re-adjustable in the field. Control shall automatically add water when needed to compensate for the water absorbed by the load. Automatic timing of each operation shall not proceed unless the proper water level and water temperature exists.
d. An indicator pilot light shall be provided to indicate when the automatic timing is proceeding.
e. A control shall be furnished to provide audible and visual means to signal the operator at the end of the laundry cycle. The signal shall persist until canceled by the operator.
f. Safety interlock device(s) shall be provided to disconnect power to the motor or motors when a door is opened while the washing machine is in operation.
g. The controls and motor(s) shall be so wired that simultaneous energizing of more than one phase of formula at a time will not occur.
h. An automatic thermal cool-down feature for processing permanent press materials shall be provided.
i. A combination thermometer-thermostat water temperature control shall be provided. The control shall provide precise temperature selections for permanent press and other fabrics. Not less than five temperature selections shall be provided with two of the selections being controlled thermostatically.

3.7.1.1.2 Automatic supply dispenser. The supply dispenser shall automatically add supplies at the correct moment, as directed by the formula control unit. The dispenser shall have not less than five supply compartments of sufficient capacity to hold supplies for one complete washing cycle. Two compartments shall be suitable for holding soap supplies. The remaining compartments shall be suitable for holding liquid supplies of bleach, blue, and sour. Design shall be such that the supply in each compartment is completely removed from the compartment when and as directed, by the formula control.

3.7.2 Moisture extractor. The moisture extractor shall be installed adjacent to the washer, and shall have the capacity compatible with the hourly output of the laundry unit. The manufacturer shall provide loading instructions for the operator. The extractor shall be capable of removing

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excess water by centrifugal force, and induce minimum vibration during high speed extraction. The extractor lid, curb, and basket shall be constructed of types 304, 304L or 316 corrosion-resistant alloy steel conforming to compositions stated in ASTM A 167. Safety features, such as an indicator for proper electrical locked lid (door cannot open during operation) and an emergency stop button shall as a minimum be provided. A watertight starter button and a factory preset timer with a manual resetable range up to 10 minutes shall be provided.

3.7.3 Vibration. Washer and extractor units shall be designed to isolate and minimize vibration transmitted to the skid and surrounding structures during the extraction cycle. The amplitude at the base of the washer and/or extractor which is in contact with the skid, shall not exceed 0.005 inch total indicator reading taken with a mass vibrometer. The washer and extractor are both to be operating in their full rated load and maximum speed configurations, with a five percent load unbalance in one pocket of each. Vibration isolation design shall not require unusual or special foundation construction materials. Anchor bolts shall be selected from readily available materials, grades and sizes to handle anticipated dynamic loads.

3.7.4 Noise. Noise level shall not exceed 85 decibels (A scale), when measured at the operators position (in front of and within 36 inches of the control panel).

3.7.5 Drying and conditioning tumbler. The general requirements of the oil-fired drying and conditioning tumbler shall conform to OO-D-1297, except the water evaporation rate shall be not less than 0.67 lb (0.3 kg) per minute, and with detail requirements as specified herein. The size and capacity of the tumbler shall be compatible with the overall performance of the laundry unit, as specified in 3.8. Controls shall be provided for processing of permanent press materials. The timing devices shall provide an adjustable range of not less than 30 minutes for drying cycle and 15 minutes for cool-down cycle. A 20-foot (6.1 m) length of flexible duct with clamps for the dryer exhaust air system shall be provided. The duct shall be capable of being routed in a manner so as not to be in the operator’s path during operation.

3.7.5.1 Burner. The burner shall be oil fired and equipped with safety devices as specified in OO-D-1297, and shall be provided with controls to maintain an adjustable preset stack operating temperature variable to not greater than 225 degrees Fahrenheit (°F) (107.2 degrees centigrade (°C)). An exhaust system shall be furnished in accordance with the manufacturer’s current practice. The exhaust system furnished shall be compatible with the water heaters exhaust system when a common system is provided. When a common system is not provided, the burner shall be provided with not less than a six foot exhaust extension.

3.7.6 Transportation brace. A removable transportation brace shall be used to support the dryer tumbler drum during transport of the laundry unit. Caution signs shall be provided to ensure that an operator does not start the dryer with the transportation brace in place. The transportation brace shall be capable of being installed or removed in five minutes by one adult person using simple hand tools. Stowage for the transportation brace shall be provided in the dryer plenum.

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3.7.7 Water heater. The water heater shall be oil fired, storage type with Type IX, stainless steel tank conforming to MIL-H-16451. Size of the water heater shall be compatible with the overall performance of the laundry unit.

3.7.7.1 Accessories. The following accessories shall be furnished with the heater:

a. One each downdraft cap.
b. One each storm collar.
c. One each elbow 90 degrees, adjustable for smoke pipe.
d. Three sections of smoke pipe, chimney or vent connector for heater.
e. One each draft regulator, barometric for smoke pipe.
f. One set of insulating couplings for inlet and outlet water connections.

3.7.8 Water supply system. A pressurized demand type, non-recirculating water supply system which includes a pump shall be provided within the laundry unit. The system shall automatically provide adequate water flow and maintain system pressure to the washer and water heater, after the connection from outside water source. Means shall be provided to prevent back flow of waste water from the laundry unit to the water source. The design of the water system shall be compatible with the overall performance of the laundry unit.

3.7.9 Storage rack(s). Adequately sized corrosion-resistant steel rack(s) with drainage for temporary storage of wet clothing shall be furnished and properly located as required. Means shall be provided to prevent discharged water from drainage system backup to the rack(s). The rack(s) shall be installed with the upper rim of the rack 32 inches ±1 inch (81.28 centimeters (cm) ± 2.5 cm) above the top surface of the skid.

3.7.10 Piping. All piping, fittings, valves, strainers, traps, and associated plumbing devices and appurtenances, shall be properly connected, and installed complete with outside connections. Piping shall be adequately sized for the intended purpose and continuous operation of the laundry unit. All piping shall be installed so as to comply with accepted plumbing practices. Piping between vibrating parts shall be protected by flexible hose connections, if the vibration is sufficient to induce leakage in rigidly connected joints. Straight run clean-outs shall be provided where required. Piping shall slope, where practicable, to inlet or outlet connections to eliminate air pockets and to permit complete drainage of the laundry unit. Fuel oil and water inlet supply connections shall be supplied with shutoff valves. Supply water inlet connection shall terminate with standard pipe threads conforming to FED-STD-H28/7. The laundry unit shall be provided with a single waste line outlet accommodating the washer, extractor, and storage rack(s) waste water. Waste line outlet connection shall terminate with fire hose threads conforming to FED-STD-H28/10, and a 20 foot length of fire hose shall be provided. Where required, piping shall include combination vents and overflow connection. All water supply and waste water lines shall be capable of being routed so as not to be in the operator’s path during operation.

3.7.11 Skid base. All components for the laundry unit shall be mounted and contained on a 7.0 foot (2.13 m) by 9.5 foot (2.89 m) welded, structural steel skid base. The base shall consist of not less than two steel skid runners with an integrally formed, welded, or bolted base plate for

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mounting components of the unit. Those areas of the skid base deck, not covered by components which require the operator to stand on during operation, shall be covered with steel bar type grating. The skid runners shall be furnished with mounting holes for anchor bolts. The number and size of holes shall be as determined by the manufacturer; to be required for anchorage of the unit under the most severe condition of vibration or thrust likely to develop during operation. The skid runners shall be provided with pockets located on the 9.5 foot (2.89 m) sides to permit lifting of the unit by a forklift truck. The pockets shall be not less than 4 inches (10.16 cm) in height by 12 inches (30.48 cm) in width located on centers of not less than 45 inches (114.3 cm) or more than 60 inches (152.4 cm), and capable of accommodating tines up to 72 inches (182.9 cm) in length. Individual mountings shall be furnished as required on the base plate for supporting motors, pumps, and other components in vertical alignment. Other components requiring removal for service, adjustment, or horizontal alignment shall be mounted in a manner to permit such removal or alignment. Tiedown devices and lifting attachments (see 3.14) shall be welded to the skid base.

3.8 Performance.

3.8.1 Laundry plant. The complete assembled laundry unit shall be capable of continuously washing and drying miscellaneous cotton clothing, at a rate of not less than 100 lb (dry weight) per hour.

3.8.2 Washing and drying of cotton clothing. Using cotton clothing as laundry; the equipment shall wash and dry the laundry to within 5 percent of its dry weight. When tested as specified in 4.4.2., cotton clothing shall be defined as “dry,” when normal dry weight is increased by no more than 7 percent.

3.8.3 Permanent press cycle. The laundry unit shall have a permanent press cycle for permanent press fabrics.

3.9 Electromagnetic interference suppression. When specified (see 6.2), the laundry unit shall be capable of meeting the UM05 requirements of MIL-STD-461 for Class 3, Group I equipment.

3.10 Lubrication. Means for lubrication shall be in accordance with the manufacturer’s standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery, with type of lubricant specified in the operator’s manual, and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.11 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or

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bolts of not less than 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be "LAUNDRY UNIT, SKID MOUNTED, 100 /HR CAPACITY."

3.12 Instruction plates. The laundry unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts, of not less than 1/8-inch (3.175 millimeters (mm)) diameter.

3.13 Cleaning, treatment and painting. Surfaces of the laundry unit normally painted shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2) and color shall conform to FED-STD-595. The end item, allied equipment, and attachments shall be the same color. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes, and the total dry film thickness shall be not less than 2.5 mils (0.06 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.14 Lifting and tiedown attachments. The laundry unit shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the laundry unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the laundry unit on the carrier when shipped.

3.15 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product. The form, fit, and function requirements of this specifications must still be satisfied.

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.16.2 Bolted connections. Boltholes shall be punched or drilled within manufacturer's design tolerances, and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be punched or drilled within manufacturer's design tolerances, and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.5 Subassemblies. Subassemblies shall be within manufacturing tolerances to permit proper fitting. Shims, spacers, caulking compounds, and weld build-up, shall not used to make allowance for poor fitting assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification, shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance ,comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein, and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete laundry unit when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory. The item must meet the requirements of the specification and be representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.1, and the packaging inspection of 4.5.

4.3 Examination. Each laundry unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall receive the tests of 4.4.1 through 4.4.2. Each production unit shall receive the test of 4.4.1. Failure to pass any test shall constitute cause for rejection.

4.4.1 Operational tests. Each laundry unit shall be completely assembled and connected to the applicable electric, water, drainage, and fuel oil services. The unit shall be operated without garments or laundry to determine that the assembly of components is complete and of proper interrelationship. The moving parts and laundry unit will show no undue vibration or overheating of bearings, and all controls will operate smoothly and positively. During the operational testing, determination shall be made to establish compliance with all performance requirements specified in section 3. The performance requirements shall be applicable to the fully assembled laundry unit, except those that require performance testing; such as meeting the capacity requirement of laundry processing through the components. The laundry unit shall be operated not less than two complete cycles, to insure that it will reliably meet the operational requirement of the specification.

4.4.2 Performance tests. The laundry unit shall be subjected to a 40-hour endurance performance test. Planned stops shall be made at 8-hour intervals during the performance test for inspections, servicing and adjustments. Records shall be kept of any adjustments of a minor nature. Major adjustments shall be cause for rejection. The laundry unit shall be tested by making all necessary connections, such as electric power, water, drainage, and fuel oil, and operated under full service

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load with applicable laundry supplies as required in the applicable paragraph of this specification. The average hourly output capacity (calculated at the end of the test) of the laundry unit shall meet the requirements as specified in 3.8. The total 40 hours shall be accumulated in 8-hour runs in accordance with the manufacturer's operating procedures. The time for each 8-hour run shall begin when appropriate laundry loads are ready for both the washer and dryer. Failure to meet the performance requirements of this specification shall be cause for rejection.

4.4.2.1 Transportation test. Prior to any operational or performance testing, a completely assembled laundry unit shall be loaded on a truck and transported 50 miles (80.5 kilometers (km)) at an average speed between 45 and 55 miles per hour (MPH) (72 and 88.5 km per hour) over improved roads and 50 miles (80.5 km) at an average speed between 20 and 30 MPH (32 and 48 km per hour) over unimproved, native soil roads. An unimproved, native soil road is an unpaved, unstabilized road to which no foreign fill material or aggregate has been added. The unimproved road shall have an undulating surface with occasional chuckholes and exposed rocks. The truck used for transport shall not have special suspension, such as air shock absorbers, for fragile loads.

4.4.2.2 Lifting and tiedown attachment tests. Lifting and tiedown attachments shall be tested for conformance to 3.14.

4.4.3 Electromagnetic interference suppression. When electromagnetic interference suppression is required (see 3.9), the laundry unit shall be tested to meet the requirements of 3.9 in accordance with MIL-STD-462.

4.5 Inspection of packaging. The sampling and inspection of the preservation of the hot water heater shall be in accordance with MIL-B-3180. The packaging and marking of the complete laundry unit, including the hot water heater, shall be in accordance with ASTM D 3951 unless otherwise stated in the contract, or by the contracting officer.

5. PRESERVATION AND PACKING

5.1 Preservation, packing, and marking. The level of preservation, packing and marking shall be in accordance with ASTM D 3951 unless otherwise specified in the contract or order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Skid-mounted laundries are used in field and semi-permanent locations, where heavy-duty laundering is required.

6.2 Suppression data. This specification supersedes MIL-L-29223A, dated 31 January 1989.

6.3 Acquisition requirements. Acquisition documents should specify the following:

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a. Number, title, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2, and 2.2).
c. When first article and inspection sample is required (see 3.3 and 4.2.1).
d. When electromagnetic interference suppression and testing is required (see 3.9 and 4.4.3).
e. Color of finish coat (see 3.13).
f. Level of preservation and level of packing (see 5.1).

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors, whether the item should be a first article sample or a standard production item from the contractor's current inventory, and the number of items to be tested (as specified in 4.2.1.). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4.1 Disposition of first article sample. Upon acceptance of the first article, the first article should remain at the manufacturing facility as a production sample, and should be the last laundry delivered on the contract. The first article should be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor should maintain the first article in a serviceable condition for the duration of the contract.

6.5 Improved design and material. It is not the intent of this specification to prohibit the procurement of laundry units incorporating newly developed designs and materials having greater reliability, less maintenance and longer life than those specified herein. However, the new designs and materials must have been evaluated and approved for use in laundry units.

6.6 Part or Identifying Number (PIN). The PIN to be used for laundry units acquired to this specification are created as follows:

OOL2880 - Specification No.

6.6.1 Part Number. The PIN is for Government purposes and does not constitute a requirement for the contractor.

6.7 Subject term (key word) listing.

Extractor
Laundry unit

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Oil-fired dryer
Open end washer
Skid-mounted

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Navy - YD1
Air Force - 99
Army - GL

Review Activity:
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

Preparing Activity:
Navy-YD1

(Project 3510-0346)

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